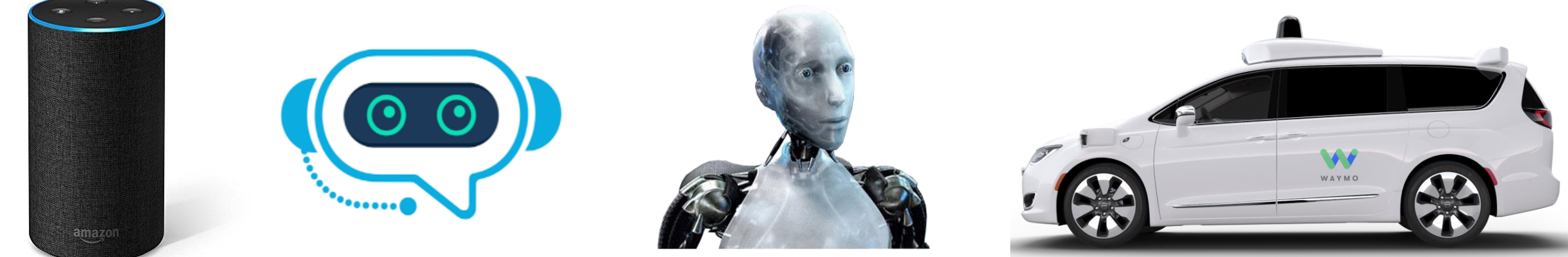


Introduction

Can a non-expert determine if an AI agent is reliable/safe for a task?



- Objective: Learn an interpretable model of a non-stationary black-box agent by interrogating it.
- Key technical challenge:
 - Which sequence of questions to ask?

Abstraction in Space of Models

Abstracted model
 (:action load_truck
 :parameters (?package ?truck ?location)
 :precondition (and (at ?truck ?location)
 (+/-/∅) (at ?package ?location))
 :effect (and (not (at ?package ?location))
 (in ?package ?truck)))

Abstracted model

This predicate can appear in three forms:
 • positive
 • negative
 • absent

abstraction ↑

Concrete model
 (:action load_truck
 :parameters (?package ?truck ?location)
 :precondition (and (at ?truck ?location)
 (at ?package ?location))
 :effect (and (not (at ?package ?location))
 (in ?package ?truck)))

Concrete model

Algorithm

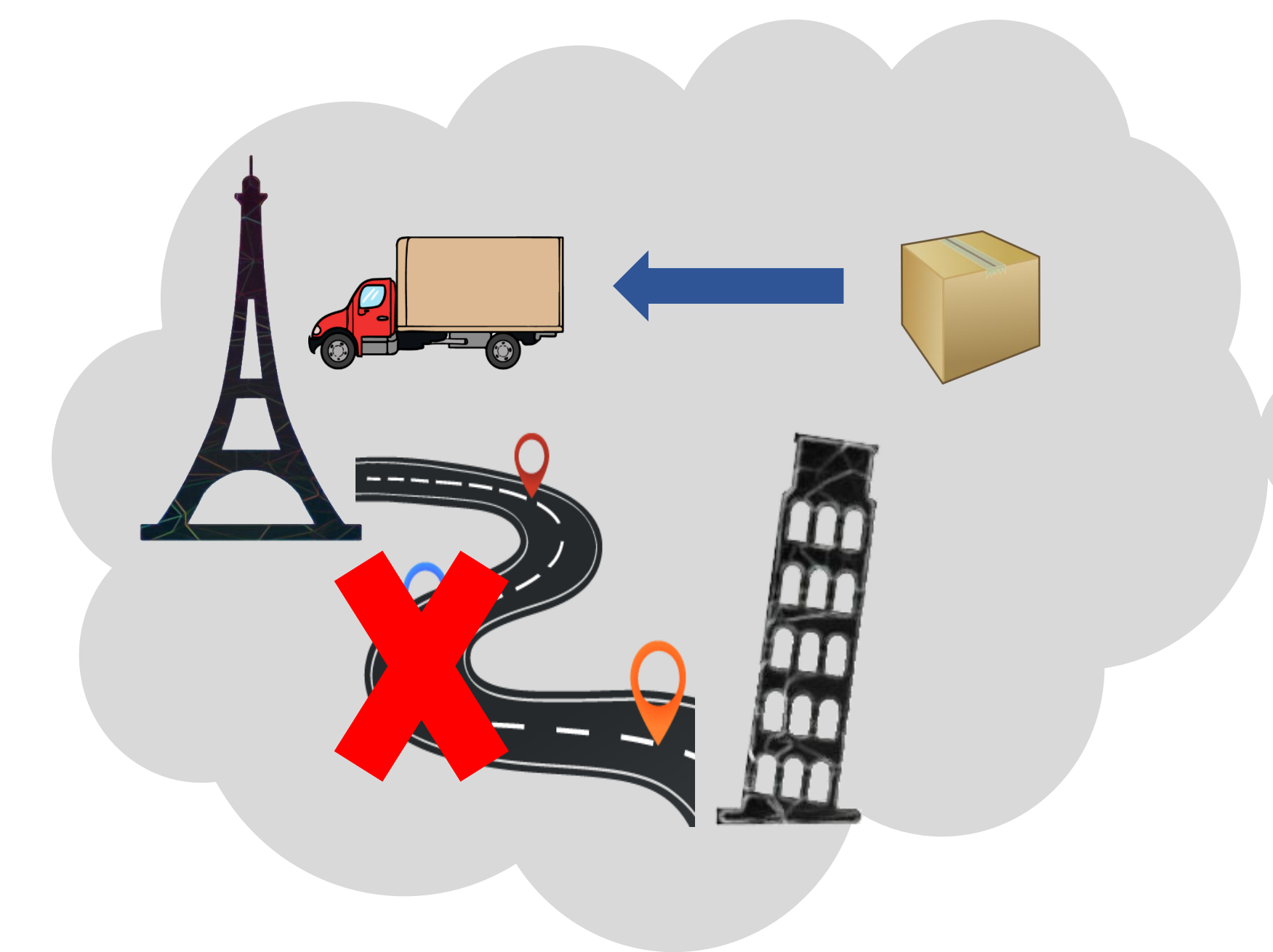
- 1 Start with the most abstracted node in lattice.
- 2 Pick abstraction candidates in some order.
- 3 For each candidate, generate three models and for each pair of models:
 - 4 • Generate a distinguishing query Q and pose it to the agent.
 - 5 • Get the response R from the agent.
 - 6 • Prune out the incorrect variants of candidate models.
 - 7 • Repeat steps 3-6 till the model is fully estimated.
 - 8 Return the final set of model(s).

Example of Agent Interrogation

Plan Outcome Query: Asks the outcome of a plan.

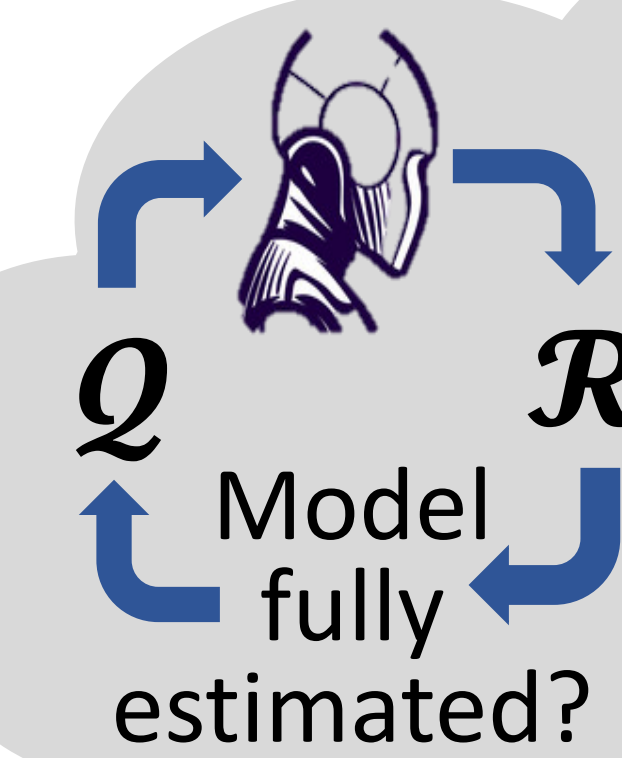
Query: Initial state, plan.

Response: Length of successful execution, final state.



What do you think will happen if you execute the plan π : $\langle \text{load_truck} (p1, t1), \text{drive} (t1, \text{Paris}, \text{Pisa}) \rangle$ starting in an initial state s_I : $(\text{at} (t1, \text{Paris}) \wedge \text{at} (p1, \text{Paris}))$?

I can execute only the first step, and the final state after executing one step was s_F : $(\neg \text{at} (p1, \text{Paris}) \wedge \text{in} (t1, p1) \wedge \text{at} (t1, \text{Paris}))$



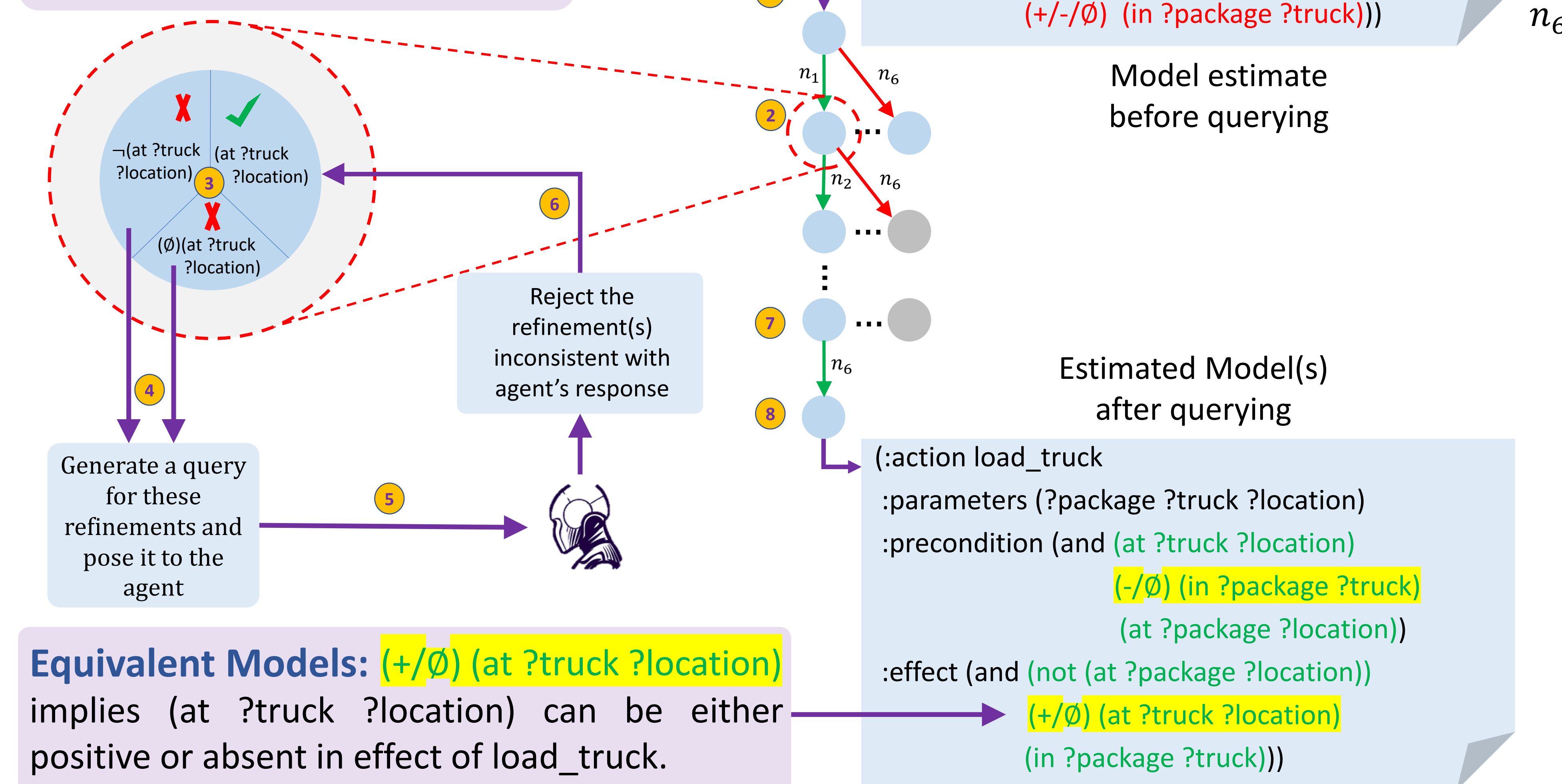
Let's keep asking queries till the model is fully estimated

I think it cannot drive between different cities.

Key Algorithmic Principle

Key feature of the algorithm

Each time we prune an abstracted model, we prune a very large number of models at the most concrete node.



Results

- Randomly generate an agent and environment from the IPC benchmark suite.
- Algorithm learns this agent's model.

Theorem: The algorithm will always terminate and return a set of models, each of which are functionally equivalent to agent's model.

Domain	P	A	$ Q_{naive} $	$ \mathcal{M}_{equiv} $	$ \mathcal{M} $	$ Q $	Time/ Q (sec)
gripper	5	3	15×2^5	1	1	37	0.14
blocks-world	9	4	36×2^9	1	1	92	1.73
elevator	10	4	40×2^{10}	64	1	109	5.91
logistics	11	6	66×2^{11}	64	1	98	11.62
parking	18	4	72×2^{18}	32	1	173	12.01
satellite	17	5	85×2^{17}	4096	1	127	19.53

Number of queries ($|Q|$) and number of models ($|\mathcal{M}|$) generated in our approach vs naive baseline. Results are averages of 10 random runs.

Salient Features

- Needs no prior knowledge of the agent model.
- Requires an agent to have only rudimentary query answering capabilities.
- Queries can be answered by the agent using a simulator.
- Works for non-stationary environments.